

Garrett Fincke

Pittsburgh, PA | garrettfinke@gmail.com | fincke.dev | github.com/ggfincke | linkedin.com/in/garrett-fincke

Education

University of Pittsburgh

Master of Science in Computer Science; SCI Scholarship for Excellence

Aug 2026 – Expected May 2028

Pittsburgh, PA

The Pennsylvania State University

Bachelor of Science in Computer Science

Aug 2021 – Dec 2024

University Park, PA

Experience

Westinghouse Electric Company

Software Developer Intern, GIC Tools and Applications

May 2026 – Present

Cranberry Township, PA

- Built a Python 3.14/FastAPI service with 15 OpenAPI-documented operations, giving an internal Databricks agent platform source-grounded access to Oracle document-index data
- Added deterministic JSON responses, request tracing, server-side filtering, hierarchy traversal, and reference resolution to make agent retrieval reproducible and debuggable
- Implemented Active Directory and Oracle authentication, rate limiting, failed-login throttling, and readiness checks; validated across 500+ baselines/releases, 1,700+ locations, and 60,000 configuration-item memberships

Scale AI

Software Engineer (contract)

May 2024 – Jul 2025

Remote

- Built Python and Docker evaluation harnesses for LLM-generated code, automating compilation, execution, linting, testing, pass@k scoring, and failure analysis
- Created schema-versioned Pydantic and JSON Schema pipelines with deterministic sampling, replacing ad hoc manual review with reproducible validation and reducing evaluation turnaround time
- Built internal dashboards for evaluation failure analysis and PostgreSQL ingest checks for schema normalization and deduplication, improving data quality across annotation and QA workflows

Projects

SwimMate v2 | *Python, Django REST, PostgreSQL/pgvector, OpenAI, Swift, SwiftUI, iOS/watchOS*

Full-stack swimming platform with native iOS/watchOS apps and AI-generated workouts

- Built a Django/PostgreSQL RAG pipeline for personalized swim workouts with hybrid retrieval and reranking; added a deterministic 48-case eval suite tracking constraint pass rates, latency, and token usage
- Developed native SwiftUI iOS/watchOS apps with HealthKit workout capture, offline-first synchronization, token refresh, and WatchConnectivity transfer

Coral | *TypeScript, React Ink, Ollama, Node.js, SQLite, TypeScript Language Server*

Local-first CLI/TUI coding agent powered by Ollama with semantic retrieval and code intelligence

- Built a multi-turn coding-agent loop with parallel tool execution, research subagents, semantic code search, MCP integration, and file, shell, search, and Git operations
- Improved local-model reliability through tool-call recovery, schema validation, loop detection, persistent SQLite sessions, conversation compaction, and TypeScript language intelligence

MDX Preview for VS Code | *TypeScript, React 19, MDX 3, VS Code Extension API*

Published VS Code/Open VSX extension for live MDX previews with 11,000+ installs

- Architected a six-package extension monorepo and extracted reusable compiler/runtime tooling into the separately published mdx-forge npm package
- Built trust-gated Safe and Trusted rendering with DOMPurify, strict CSP, React component support, and compatibility shims for five MDX frameworks

Technical Skills

Languages: Python, TypeScript/JavaScript, Swift, Java, SQL, C/C++

Backend & Data: Django, FastAPI, Node.js, PostgreSQL/pgvector, Oracle, SQLite, Pydantic, REST/OpenAPI

AI/LLM: RAG, embeddings/vector search, agentic AI, MCP, Ollama, OpenAI/Anthropic APIs

Frontend & Mobile: React, Next.js, SwiftUI, UIKit, Tailwind CSS

Cloud & DevOps: Docker, Docker Compose, AWS EC2/SSM/CloudWatch, GitHub Actions, CI/CD, Vercel